DATA ITEM DESCRIPTION

Title: Model-Based Quality Assurance - Data Framework (MBQA-DF) for First Article Inspection Report and Certificate of Conformance

Number: DI-SESS-82558
AMSC Number: 10698
DTIC Applicable: No
Preparing Activity: 11 (AFLCMC/EZS)
Applicable Forms: None

Approval Date: 20260917
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2026-039

Use / Relationship:

This Data Item Description (DID) defines the format, content, and application of the Model-Based Quality Assurance – Data Framework (MBQA-DF). It aligns with SAE International standards AS9102 First Article Inspection Report (FAIR) and AS9163 Certificate of Conformity (CoC) (https://www.sae.org/standards) to standardize digital quality data for contractual deliverables and digital thread integration.

The MBQA-DF contributes to a broader digital thread by enabling integration of FAIR and CoC elements into the overarching digital ecosystem. It ensures quality assurance data is digitally interoperable with Product Lifecycle Management (PLM) systems, digital thread frameworks and other lifecycle systems as part of a unified approach to design, manufacturing, and quality processes. The MBQA-DF provides a structured, model-based framework that supports consistent and traceable development of FAIR and CoC data artifacts, ensuring alignment with program requirements and improving integration across digital engineering environments. FAIR references CoC data (e.g., material specifications, conformance status), and CoC references FAIR information (e.g., inspection results and compliance status). This interrelationship ensures that quality assurance data can be delivered either as combined or separate items, while maintaining data integrity and usability.

The MBQA-DF is aligned with, and interfaces with, DMSC MBC 1.0 – 2024 (https://qifstandards.org/model-based-characteristics-standard/) and ISO 23952 (https://www.iso.org/standards.html) to provide standardized, cross-domain integration and persistent identification of model-based quality characteristics. These connections ensure a unified approach to design, manufacturing, and quality processes within the digital ecosystem.

This DID specifies only the required data outputs for contract delivery and does not address the execution of tasks described in SAE International standards. The official and complete references for the First Article Inspection Report (FAIR) and Certificate of Conformance (CoC) standards are found in the full versions of SAE AS9102 and SAE AS9163, as published under copyright by SAE International.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.
Source: https://assist.dla.mil -- Downloaded: 2026-10-02T19:35Z
Check the source to verify that this is the current version before use.

This DID is the product of a pilot effort to model a DID using System Modeling Language (SysML). SysML is a general-purpose architecture modeling language for Systems Engineering applications. SysML supports the specification, analysis design, verification, and validation of a broad range of systems and systems-of-system. These systems may include hardware, software, information, processes, personnel, and SysML is a dialect of UML 2, and is defined as a UML 2 Profile. A UML Profile is a UML dialect that customizes via three main mechanisms: Stereotypes, Tagged Values and Constraints. SysML is an enabling technology for Systems Engineering.

The attached digital reference profile(s) does not constitute DoW endorsement of a commercial tool or vendor. It is provided as a reference implementation to facilitate the testing of ASSIST and the future revision to DSP procedures to accommodate digital standardization products.

Further instructions for how to implement, model, and customize the MBQA-DF Profile and Reference Architecture can be found in the model itself and the attached Guidebook for the Implementation of MBQA-DF.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

2. Format. The MBQA-DF shall be delivered in the native file format(s) of the modeling tool(s), such as an mdzip file, or developed directly within the Integrated Digital Data Environment (IDDE) identified in the Contractor's Systems Engineering Management Plan (SEMP). It shall meet the requirements of the attached Profile-Viewpoint Overview, following prescribed stereotypes, attributes, and structure that enable the generation of required views.

3. Content. The MBQA-DF shall:
   - 3.1. Conform to the design views and data elements specified in the attached Profile-Viewpoint Overview.
   - 3.2. Follow the structure and attributes defined in the attached MBQA-DF Model Structure, ensuring consistency with the stereotypes and attributes required by the contract.
   - 3.3. Be capable of producing the required views defined in the attached Profile-Viewpoint Overview.
   - 3.4. Maintain alignment with the structured data representations described in the attached MBQA-DF Model Structure, ensuring traceability between model-based data elements and the digital thread components required for FAIR and CoC.
   - 3.5. Support customization based on contract-specific requirements, such that any views not applicable to the SOW or excluded by the approved CDRL can be removed or adjusted to meet contract-specific guidance without altering the core MBQA-DF structure.
   - 3.6. Attached documents.
     a. MBQA-DF Guidebook (MSWord file): An aid for implementing the requirements of this DID to create the machine-readable First Article Inspection Report (FAIR) and Certificate of Conformity (CoC).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.
Source: https://assist.dla.mil -- Downloaded: 2026-10-02T10:35Z
Check the source to verify that this is the current version before use.

b. MBQA-DF (Digital reference profile set of 3-mdzip files): The model containing: 1) a Profile file that provides the schema, standards, and relationships required to implement in the digital thread, 2) a Model Structure file that defines the hierarchical organization and domain-specific data necessary for model-based quality analyses, and 3) an Ontology file showing how the quality data is interoperable across various digital thread domains, whether originating from model-based tools or traditional legacy systems.
c. MBQA-DF Profile-Views v1 (MSExcel file): Defines the required design views and data elements for the MBQA-DF FAIR and CoC content, respectively.

3.7. Saving Attached Files. After opening the PDF file, please ensure you have selected "Navigation View" to view the attached files.
a. Use "Save-As" from Adobe and save each file to your selected folder.
b. You can now launch the application from your file folder.
c. When launching a selected file, you must open the file in its respective application

End of DI-SESS-82558.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:04Z
Check the source to verify that this is the current version before use.
Source: https://assist.dla.mil -- Downloaded: 2026-10-02T19:35Z
Check the source to verify that this is the current version before use.